

All Examples with Solutions

Chapter 3: Diode Models and Circuits

Examples 3.1 – 3.35

Contents

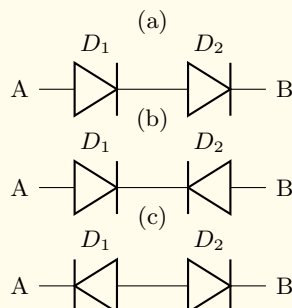
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1 Ideal Diode Examples (3.1 – 3.9)

Example 3.1 — Series Combinations of Diodes (Fig. 3.4)

Problem: Three configurations of two series diodes. Determine which can conduct current.

Circuit Diagrams



Solution

- **(a) Same direction:** Assume both ON. Current flows $A \rightarrow B$ through both anodes $\rightarrow$ cathodes. $I_D > 0$ for both. ✓ **CONDUCTS.**
- **(b) Cathodes meeting:** Try D_1 ON (current $A \rightarrow$ center). Then D_2 needs current center $\rightarrow B$, but that's cathode $\rightarrow$ anode for D_2 . $I_{D2} < 0 \Rightarrow$ **CONTRADICTION.**
Why it fails: Two one-way valves facing each other. No matter which direction current flows, one valve blocks.
- **(c) Anodes meeting:** Same symmetric deadlock as (b). **BLOCKED.**

Result: Only (a) conducts — both diodes must point the same direction.

Example 3.2 — I/V of an Ideal Diode (Fig. 3.5)

Q: The I/V characteristic doesn't show I_D for $V_D > 0$?

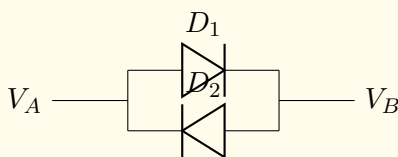
Solution

The vertical line at $V_D = 0$ represents *any* current from 0 to ∞ . In a real circuit with finite sources, the external circuit limits the current. The diode simply sustains $V_D = 0$ (short circuit behavior).

Example 3.3 — Antiparallel Diodes (Fig. 3.6a)

Problem: Plot I/V for $D_1 \parallel D_2$ connected antiparallel.

Circuit



Solution

For $V_A > 0$: D_1 is forward-biased $\Rightarrow$ short $\Rightarrow I_A$ unlimited.

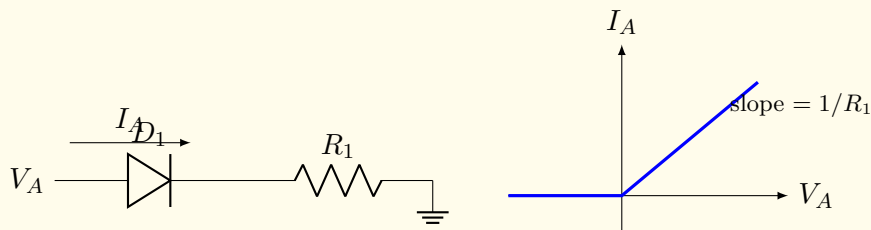
For $V_A < 0$: D_2 is forward-biased $\Rightarrow$ short $\Rightarrow I_A$ unlimited.

Acts as a **short circuit for ALL voltages**. The I/V is the entire vertical axis.

(With real diodes and the CV model, this becomes useful for limiting circuits in Section 3.5.3.)

Example 3.4 — Diode-Resistor I/V Characteristic (Fig. 3.7)

Problem: Plot I/V for D_1 in series with R_1 .

Circuit and I/V Plot**Solution — Assume-and-Verify**

Case 1: $V_A < 0$

1. Assume D_1 OFF (open circuit)
2. $I_A = 0$ (no current path)
3. Verify: $V_D = V_A - 0 = V_A < 0$ ✓ (reverse biased)

Case 2: $V_A > 0$

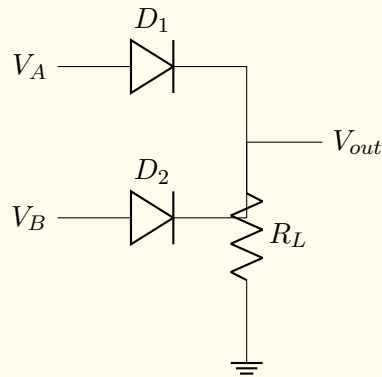
1. Assume D_1 OFF $\Rightarrow I_A = 0$, no drop across $R_1 \Rightarrow V_D = V_A > 0$
2. **CONTRADICTION:** $V_D > 0$ means diode should be ON
3. Re-assume D_1 ON (short): $I_A = V_A/R_1 > 0$ ✓

$$I_A = \begin{cases} 0 & V_A < 0 \\ V_A/R_1 & V_A \geq 0 \end{cases}$$

Why: Negative V_A pushes current cathode $\rightarrow$ anode — blocked. Positive V_A pushes anode $\rightarrow$ cathode — allowed.

Example 3.5 — Why I/V not V/I?**Solution**

Voltage polarities are predicted more easily than current polarities. Also, transistors produce current in response to voltage. Convention uses I/V.

Example 3.6 — Diode OR Gate (Fig. 3.8)**Problem:** $V_A, V_B \in \{0, +3\text{ V}\}$. Find V_{out} .**Circuit****Solution**

If $V_A = 3\text{ V}$, $V_B = 0\text{ V}$: D_1 's anode at 3 V , cathode at V_{out} . D_1 ON $\Rightarrow V_{out} = 3\text{ V}$. Now D_2 : anode at 0 V , cathode at $3\text{ V} \Rightarrow V_D = -3\text{ V} \Rightarrow D_2$ OFF $\checkmark$.
 By symmetry: $V_{out} = \max(V_A, V_B)$ — this is a logical **OR gate**.

Example 3.7 — Diode at $V_D = 0$ **Solution**

State is ambiguous: $I_D = 0$ and $V_D = 0$ simultaneously. Diode is at the exact boundary. Apply a tiny perturbation to resolve.

Example 3.8 — Characteristic Similarity**Solution**

Figs. 3.7(d) and 3.10(d) look alike because V_{out} in the rectifier equals $I_A \cdot R_1$ in the diode-resistor circuit. They differ by the scaling factor R_1 .

Example 3.9 — Signal Strength Indicator**Problem:** 1.8 GHz signal with peak from $2\text{ }\mu\text{V}$ to 10 mV applied to a rectifier.**Solution**

Using $V_{avg} = V_p/\pi$:

$$\text{Min: } \frac{2\text{ }\mu\text{V}}{\pi} = 0.637\text{ }\mu\text{V} \quad \text{Max: } \frac{10\text{ mV}}{\pi} = 3.18\text{ mV}$$

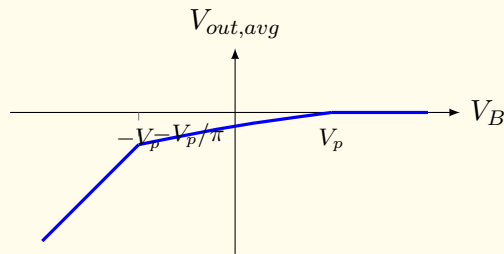
The DC average is proportional to signal strength — hence “signal strength indicator.”

2 Limiter & CV Model Examples (3.10 – 3.16)

Example 3.10 — Average Voltage vs. Battery Sweep (Fig. 3.11c)

Problem: Sketch $V_{out,avg}$ as battery voltage V_B sweeps from $-\infty$ to $+\infty$.

Sketch of $V_{out,avg}$ vs. V_B



Solution

- $V_B < -V_p$: D_1 always ON, output clamped to $V_B \Rightarrow$ average = V_B (linear ramp)
- $-V_p < V_B < 0$: partial clipping $\Rightarrow$ average between V_B and $-V_p/\pi$
- $V_B = 0$: same as Fig. 3.11(b) $\Rightarrow$ average = $-V_p/\pi$
- $V_B \geq V_p$: D_1 never turns on $\Rightarrow V_{out} = V_{in}$, average = 0 (flat)

Example 3.11 — Is Fig. 3.11(b) a Rectifier?

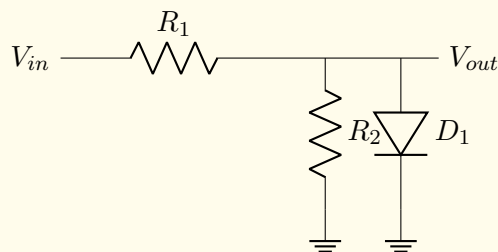
Solution

Yes. It passes only *negative* cycles, producing a negative average. To pass positive cycles instead, swap the diode terminals.

Example 3.12 — Voltage Divider with Diode (Fig. 3.14)

Problem: Plot I/O characteristic using (a) ideal and (b) CV model.

Circuit



Solution**(a) Ideal Model:**

- $V_{in} < 0$: D_1 OFF $\Rightarrow V_{out} = V_{in}$ (slope 1, passes through)
- $V_{in} > 0$: D_1 ON (short) $\Rightarrow R_2 \parallel 0 = 0$, so

$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in} \quad (3.9)$$

(b) Constant-Voltage Model:

- $V_{in} < V_{D,on}$: D_1 OFF $\Rightarrow V_{out} = V_{in}$ (slope 1)
- $V_{in} \geq V_{D,on}$: D_1 ON as $V_{D,on}$ battery. By KCL:

$$\frac{V_{in} - V_{out}}{R_1} = \frac{V_{out}}{R_2} + \frac{V_{out} - V_{D,on}}{0^+}$$

$$V_{out} = \frac{R_2 V_{in} + R_1 V_{D,on}}{R_1 + R_2} \quad (3.11)$$

Why breakpoint shifts: The diode needs $V_{D,on}$ across it to turn on. Since $V_{out} = V_{in}$ when OFF, the breakpoint occurs at $V_{in} = V_{D,on}$.

Example 3.13 — Parallel Diodes, Different Areas (Fig. 3.15)

Problem: $D_1 \parallel D_2$ with different I_S . Find individual currents.

Solution

Why exponential model? Ideal/CV models can't distinguish different diode areas. Since both share the same V_D : $V_T \ln(I_{D1}/I_{S1}) = V_T \ln(I_{D2}/I_{S2})$

$$\frac{I_{D1}}{I_{S1}} = \frac{I_{D2}}{I_{S2}} \quad (3.14)$$

With $I_{in} = I_{D1} + I_{D2}$:

$$I_{D1} = \frac{I_{in}}{1 + I_{S2}/I_{S1}}, \quad I_{D2} = \frac{I_{in}}{1 + I_{S1}/I_{S2}} \quad (3.15-3.16)$$

Current splits proportionally to I_S (proportional to cross-section area).

Example 3.14 — Diode Clamp (Fig. 3.16)**Solution**

D_1 OFF: $V_{out} = \frac{R_2}{R_1 + R_2} V_{in}$.

Breakpoint: $V_{out} = V_{D,on}$ requires $V_{in} = (1 + R_1/R_2) V_{D,on}$.

Why $(1 + R_1/R_2)$ factor? At the edge, D_1 draws zero current, so the circuit is a pure voltage divider. To get $V_{out} = V_{D,on}$ at the divider output, input must be amplified by the inverse of the divider ratio.

After breakpoint: $V_{out} = V_{D,on}$ (clamped).

Example 3.15 — Output Across Series Resistor (Fig. 3.17)**Solution**

D_1 ON for very negative V_{in} : $V_{out} = V_{in} + V_{D,on}$ (output across R_1).

D_1 turns off at $V_{in} = -(1 + R_1/R_2) V_{D,on}$.

After: $V_{out} = \frac{R_1}{R_1+R_2} V_{in}$.

Example 3.16 — Two-Diode Circuit (Fig. 3.18)

Problem: Circuit with D_1 , D_2 , resistors, and a 2V battery. Find V_{out} vs. V_{in} .

Solution — Three Regions

Region 1 ($V_{in} \ll 0$): Try D_1 ON, D_2 ON $\Rightarrow$ current through D_2 flows cathode $\rightarrow$ anode $\Rightarrow$ **CONTRADICTION**.

Re-assume D_1 ON, D_2 OFF:

$$V_{out} = (V_{in} + V_{D,on}) \frac{R_2}{R_1 + R_2}$$

Breakpoint 1: D_1 turns off at $V_{in} = -V_{D,on}$.

Region 2 ($-V_{D,on} < V_{in} < V_B + V_{D,on}$): Both OFF, $V_{out} = 0$.

Breakpoint 2: D_2 turns on at $V_{in} = V_B + V_{D,on}$.

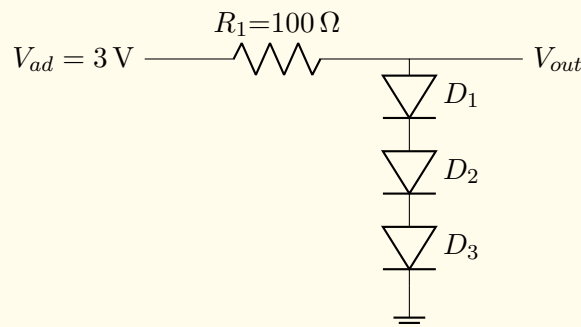
Region 3 ($V_{in} > V_B + V_{D,on}$): D_1 OFF, D_2 ON:

$$V_{out} = V_{in} - V_{D,on} - V_B$$

Why this order? As V_{in} rises from $-\infty$, D_1 provides a path first (faces forward). D_2 can't conduct until V_{in} overcomes both $V_{D,on}$ and V_B .

3 Exponential & Small-Signal Examples (3.17 – 3.22)**Example 3.17 — Cellphone Charger (Fig. 3.19)**

Problem: 3V adaptor, $R_1 = 100\Omega$, three identical diodes produce $V_{out} = 3V_D \approx 2.4\text{V}$. Find I_S . Then find V_{out} when adaptor changes to 3.1V.

Circuit

Solution**Part (a):** Finding I_S :

$$I_X = \frac{3 - 2.4}{100} = 6 \text{ mA}$$

Each diode: $V_D = 2.4/3 = 0.8 \text{ V}$.

$$6 \text{ mA} = I_S \exp \frac{0.8}{0.026} \Rightarrow \boxed{I_S = 2.6 \times 10^{-16} \text{ A}}$$

Part (b): $V_{ad} = 3.1 \text{ V}$. Iterate:

1. Guess $I_X = 7 \text{ mA} \Rightarrow V_{out} = 3V_T \ln(7\text{mA}/I_S) = 2.412 \text{ V}$
2. Check: $I_X = (3.1 - 2.412)/100 = 6.88 \text{ mA}$
3. Recompute: $V_{out} = 3V_T \ln(6.88\text{mA}/I_S) = 2.411 \text{ V} \checkmark$ converged

$$\boxed{\Delta V_{out} = 2.411 - 2.400 = 11 \text{ mV}} \text{ for } 100 \text{ mV input change!}$$

Why only 11 mV? The logarithmic V-I relationship means voltage changes very slowly with current. Three diodes act as a crude voltage regulator.

Example 3.18 — Small-Signal Quick Calculations**Setup:** Diode biased at $I_D = 1 \text{ mA}$. $r_d = 26 \text{ mV}/1 \text{ mA} = 26 \Omega$.**Solution**(a) $\Delta V_D = 1 \text{ mV}$:

$$\Delta I_D = \frac{\Delta V_D}{r_d} = \frac{1 \text{ mV}}{26 \Omega} = \boxed{38.4 \mu\text{A}}$$

(b) $\Delta I_D = 0.1 \text{ mA}$ (10% change):

$$\Delta V_D = r_d \cdot \Delta I_D = 26 \times 0.1 = \boxed{2.6 \text{ mV}}$$

Insight: 10% current change = only 2.6 mV voltage change. Confirms the CV model's validity.

Example 3.19 — Sinusoidal Perturbation on a Diode**Solution** $V(t) = V_0 + V_p \cos \omega t$ with $V_p \ll V_T$:

$$I_D(t) = I_0 \left(1 + \frac{V_p}{V_T} \cos \omega t \right), \quad I_0 = I_S \exp(V_0/V_T)$$

The diode acts as a linear resistor $r_d = V_T/I_0$ for the AC component.

Example 3.20 — Reverse Small-Signal Derivation**Solution**

Starting from $V_D = V_T \ln(I_D/I_S)$:

$$\frac{dV_D}{dI_D} = \frac{V_T}{I_D} = r_d$$

Same result as Eq. (3.58) — confirms consistency.

Example 3.21 — Small-Signal Charger Revisited (Fig. 3.25)**Solution**

Replace each diode with $r_d = 26 \text{ mV}/6 \text{ mA} = 4.33 \Omega$. Total: $3r_d = 13 \Omega$.
Small-signal voltage divider:

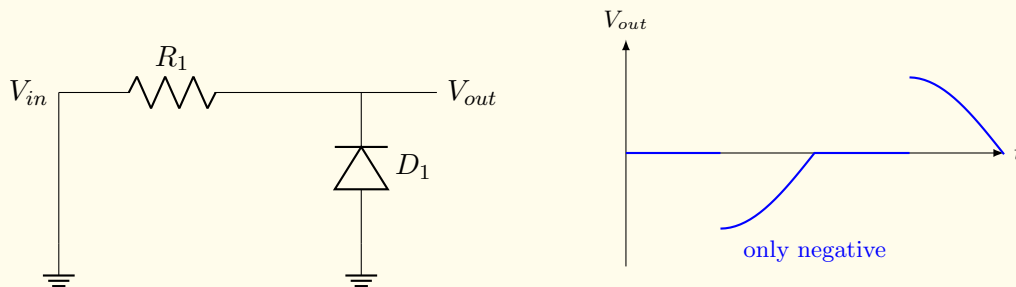
$$v_{out} = \frac{3r_d}{R_1 + 3r_d} v_{ad} = \frac{13}{113} \times 100 \text{ mV} = \boxed{11.5 \text{ mV}}$$

Compare with iterative answer of 11 mV $\Rightarrow$ excellent accuracy!

Example 3.22 — Load Effect on Charger**Solution**

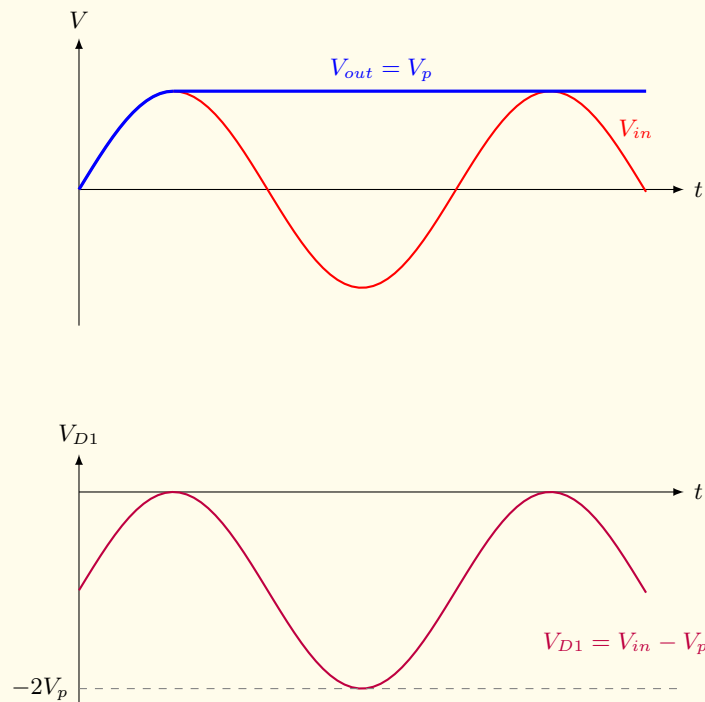
Load draws 0.5 mA $\Rightarrow$ reduces diode current by 0.5 mA.

$$\Delta V_{out} = \Delta I_D \cdot (3r_d) = 0.5 \text{ mA} \times 13 \Omega = \boxed{6.5 \text{ mV}}$$

4 Rectifier Application Examples (3.23 – 3.31)**Example 3.23 — Alternative Rectifier (Fig. 3.29)****Circuit and Output Waveform****Solution**

D_1 ON for $V_{in} \leq -V_{D,on}$. OFF for $V_{in} > -V_{D,on}$.

Blocks **positive** cycles, passes negative $\Rightarrow$ also a valid rectifier (negative output).

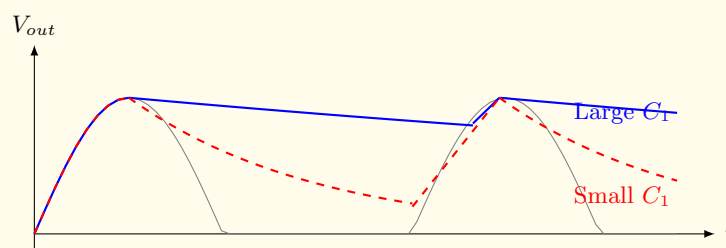
Example 3.24 — Peak Detector Analysis (Fig. 3.31)**Input, Output, and Diode Voltage Waveforms****Solution**

With ideal D_1 : C_1 charges to V_p at the first peak. After that, $V_{out} = V_p$ (constant).
 $V_{D1} = V_{in} - V_{out} = V_p \sin \omega t - V_p$. Swings from 0 (at peak) to $-2V_p$ (at negative peak).
Key insight for voltage doublers: The diode voltage looks like the original sine wave, but shifted down by V_p . Average diode voltage = $-V_p$.

Example 3.25 — Laptop Power Requirements**Solution**

$P = 25 \text{ W}$ at $V = 3.3 \text{ V}$:

$$I = \frac{P}{V} = \frac{25}{3.3} = 7.58 \text{ A}, \quad R_L = \frac{V}{I} = \frac{3.3}{7.58} = \boxed{0.436 \Omega}$$

Example 3.26 — Effect of C_1 on Ripple**Output Waveform for Different C_1 Values**

Solution

- **Large C_1 :** Slow discharge $\Rightarrow$ small $V_R \Rightarrow$ nearly DC output (blue curve)
- **Small C_1 :** Fast discharge $\Rightarrow$ approaches half-wave rectified waveform (red curve)
- $C_1 = 0$: No energy storage $\Rightarrow$ pure half-wave rectifier
- $C_1 = \infty$: Perfect DC at $V_p - V_{D,on}$

Example 3.27 — Filter Capacitor Design

Problem: $V_p = 4.5\text{ V}$, $R_L = 0.436\ \Omega$, $f = 60\text{ Hz}$, $V_R \leq 0.1\text{ V}$. Find C_1 .

Solution

Using Eq. (3.80):

$$C_1 = \frac{V_p - V_{D,on}}{V_R \cdot R_L \cdot f} = \frac{4.5 - 0.8}{0.1 \times 0.436 \times 60} = \frac{3.7}{2.616} = \boxed{1.414\text{ F}}$$

This is enormous! A 1.4 F capacitor for a laptop adaptor is impractical. This is why real adaptors use **switching regulators** (Chapter 15) instead of linear ones.

Example 3.28 — Peak Diode Current**Solution**

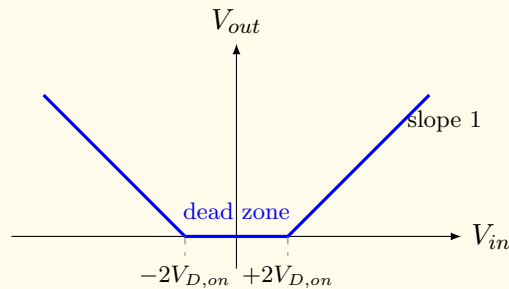
Using Eq. (3.92):

$$I_p = \frac{V_p}{R_L} \left(R_L C_1 \omega \sqrt{\frac{2V_R}{V_p} + 1} \right)$$

With $V_p = 4.5\text{ V}$, $R_L = 0.436\ \Omega$, $C_1 = 1.414\text{ F}$, $\omega = 377$, $V_R = 0.1\text{ V}$:

$$I_p = \frac{4.5}{0.436} \left(0.436 \times 1.414 \times 377 \times \sqrt{\frac{0.2}{4.5} + 1} \right) \approx \boxed{517\text{ A}}$$

Extremely large — the capacitor demands huge charging current in a very short pulse.

Example 3.29 — FW Rectifier I/O Characteristic**Transfer Characteristic**

Solution

$$V_{out} = \begin{cases} 0 & |V_{in}| < 2V_{D,on} \\ |V_{in}| - 2V_{D,on} & |V_{in}| \geq 2V_{D,on} \end{cases}$$

Slope = 1 in both directions. Dead zone of width $2V_{D,on}$ around zero.

Why $2V_{D,on}$? Two diodes are always in the current path — one on each side of the bridge.

Example 3.30 — Bridge Diode Currents (Fig. 3.41)**Solution**

Each diode pair (D_1, D_2 or D_3, D_4) alternately carries $I_L = V_{out}/R_L$ during its respective half cycle. Average current per diode = $I_L/2$.

Example 3.31 — Full-Wave Rectifier Design

Problem: Deliver 2W at 3.6V with ripple ≤ 0.2 V.

Solution

$$V_{in,p} = 3.6 + 2(0.8) = 5.2 \text{ V}$$

$$I_L = P/V = 2/3.6 = 0.556 \text{ A}$$

Using FW ripple formula:

$$C_1 = \frac{I_L}{2 \cdot V_R \cdot f} = \frac{0.556}{2 \times 0.2 \times 60} = \boxed{23,000 \mu\text{F}}$$

Still large, but much more practical than the HW case!

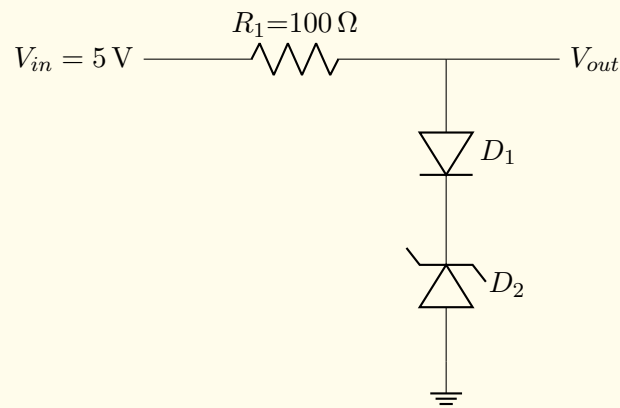
5 Regulator, Limiter & Multiplier Examples (3.32 – 3.35)**Example 3.32 — Small-Signal Rectification?****Solution**

A 10 mV signal **cannot** turn real diodes on/off (needs ~ 800 mV). Options:

- Add DC bias of $V_{D,on}$ to put the diode at the edge of conduction
- Use “precision rectification” with op-amps (Chapter 8)

Example 3.33 — Zener Regulator with Forward Diode (Fig. 3.45)

Problem: $V_{in} = 5 \text{ V}$, $R_1 = 100 \Omega$, D_1 (forward, $V_{D,on} = 0.8 \text{ V}$), D_2 (Zener, $V_{DZ} = 2.7 \text{ V}$, $r_{D2} = 5 \Omega$).

Circuit**Solution****Bias calculation:**

$$V_{out} = V_{D,on} + V_{DZ} = 0.8 + 2.7 = 3.5 \text{ V}$$

$$I_D = \frac{V_{in} - V_{out}}{R_1} = \frac{5 - 3.5}{100} = \boxed{15 \text{ mA}}$$

Small-signal parameters:

$$r_{D1} = \frac{26 \text{ mV}}{15 \text{ mA}} = 1.73 \Omega$$

Line regulation:

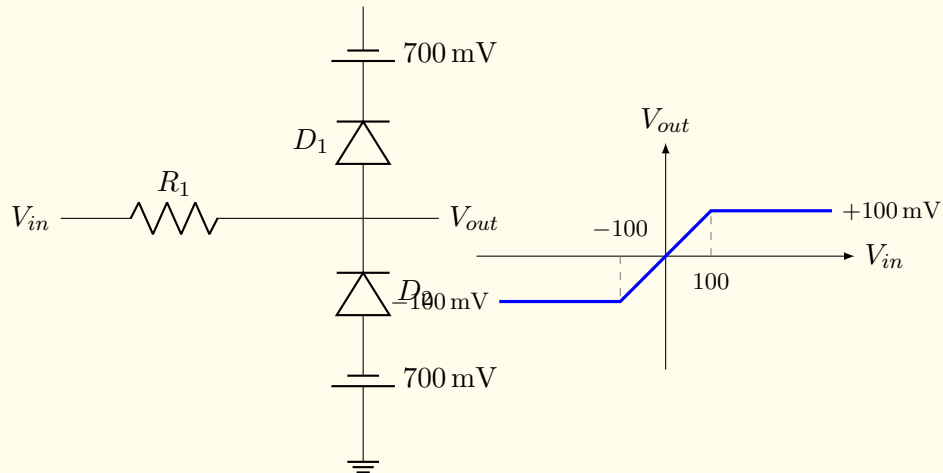
$$\frac{v_{out}}{v_{in}} = \frac{r_{D1} + r_{D2}}{r_{D1} + r_{D2} + R_1} = \frac{1.73 + 5}{106.73} = \boxed{0.063}$$

Only 6.3% of input variation reaches the output!

Load regulation:

$$\left| \frac{v_{out}}{i_L} \right| = (r_{D1} + r_{D2}) \parallel R_1 = \frac{6.73 \times 100}{106.73} = \boxed{6.31 \Omega}$$

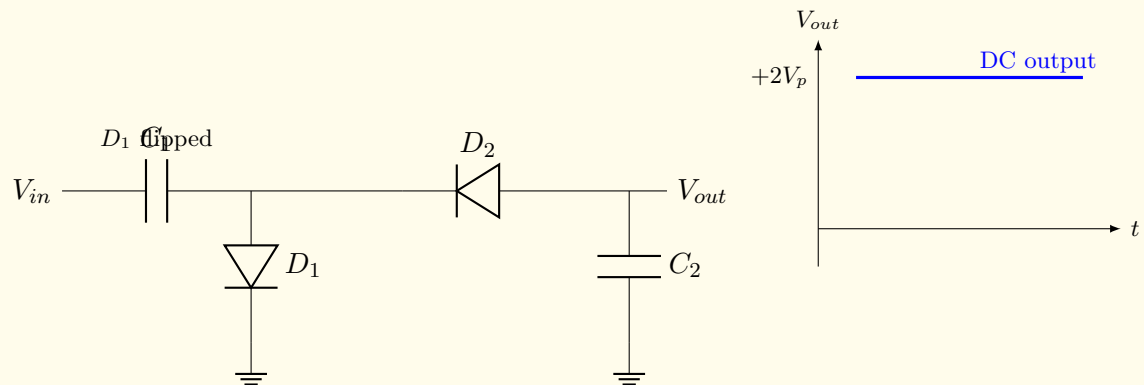
A 1 mA load change causes only 6.3 mV output change.

Example 3.34 — Limiter Design at $\pm 100\text{ mV}$ **Circuit and Transfer Characteristic****Solution**

With $V_{D,on} = 800\text{ mV}$, need $V_L = 100\text{ mV}$.

Each battery: $V_B = V_L - V_{D,on} = 100 - 800 = \boxed{-700\text{ mV}}$

Why negative? The natural clipping point is at $\pm 800\text{ mV}$. To shift it **inward** to $\pm 100\text{ mV}$, the battery must **oppose** the diode drop.

Example 3.35 — Reversed Voltage Doubler (Fig. 3.56)**Reversed Doubler Circuit and Waveforms**

Solution

With D_1 flipped:

- D_1 ON during **positive** swings (charges C_1 with opposite polarity)
- C_1 stores V_p such that the signal is shifted **downward**
- Peak of shifted signal: $-2V_p$ (but D_2 is also flipped)
- $D_2 + C_2$ capture the positive peak

$$V_{out} = +2V_p - 2V_{D,on}$$

(Same magnitude as original doubler, output is positive.)